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User

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COMP3010 Security Operations & Incident Management

Coursework 1: Set Exercises – Analysis

Task:

The Set Exercises in this module are focused on undertaking and analysing several experimental based lab exercises. Within the module plan, several laboratory exercises provide a better understanding of the capability of incident detection and management tools. These include:

• Effectiveness of different network intrusion/monitoring tools

• Gathering information for an incident report

• Exploration of pcap packets for evidence

• Investigating alerts for intruders

You work as a security analyst. One day you receive the following pcap file with an intrusion incident detected. You are required to perform a detailed intrusion analysis on the provided network traffic logs. A set of questions have been prepared to aid you in your analysis. You will find them on as a quiz on the DLE. Using the answers from the questions in the quiz (separate document), write a 1000-word report on your findings, detailing the following:

• Identify the Infected System: Detect which system in the network has been compromised.

• Analyse How the Infection Occurred: Determine the method used to infect the system.

• Identify the Type of Infection: Describe what type of malware or attack caused the infection.

Important Notes:

• This is individual work and not a textbook-based exercise.

• You are expected to conduct a practical experiment, document your process, analyse the results, and produce an incident report.
• Your report should focus on the application of knowledge rather than merely presenting theoretical or textbook information. Critical analysis and thorough documentation of your investigative process are key to this assignment.

Coursework 1 Deliverables

1. Investigation Validation Exercise Students complete investigation validation questions supplied through the DLE. These act as milestones, verification checkpoints and accuracy checks. They are not the primary assessment.
2. Investigation Report Maximum 1,000 words, excluding references, evidence appendix and AI reflection appendix. The report should communicate the investigation as a coherent professional case study rather than as a list of answers.

Recommended focus What should be demonstrated
Methodology Selection and justification of appropriate investigation methods and tools.
Analysis Systematic examination of relevant artefacts and interpretation of findings.

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Recommended focus What should be demonstrated
Findings Clear, evidence-based conclusions about the intrusion or monitoring problem.
Technical reasoning Explanation of how observations support conclusions and where uncertainty remains.
Table 3: Report sections
3. Evidence Portfolio – you may include this in a git repository and attach the link in your report.
• Screenshots and annotated tool outputs
• Packet captures and stream analysis
• IDS outputs and detection rules
• Logs, timelines and other relevant artefacts
• Use of AI logs, methodology and sample prompt/queries
• Other evidence necessary to demonstrate authentic investigation
4. Video Briefing
Maximum 5 minutes. Audience: SOC Lead, Incident Manager or Security Manager. The briefing should resemble a professional analyst briefing, focusing on the situation, key evidence, findings, implications and recommended next steps.
5. AI Reflection
Maximum 250 words. Students must critically discuss how AI was used, where it was beneficial, outputs generated, hallucinations or errors encountered, validation steps performed, and risks and

limitations of AI assistance. Any handicap in use of AI needs to be mentioned here – limited to freetier or open source.

The investigation validation exercise questions will be provided as Quiz 1 on the DLE.

Acceptable level of generative AI tool use in this coursework element
Partnered Work P1 – Generative AI tool use forms an integral part of the assessment and is compliant with the assessment brief. ☒
Table 4: Acceptable Level of generative AI use

Any use of AI in your work must be declared within your documentation. You must also include a signed Generative AI Declaration as an appendix to your submission. The declaration form can be found on DLE (or Here on the Programme Page). This form will not be included in any word count associated with this assignment.

The report must not exceed 1000 words or 10 pages (whichever shorter) excluding screenshots and references. Please note that anything exceeding the maximum number of pages will not be accessed by the markers. A Rubric will be used to assess and provide feedback. The template for this can be found below (Table 2).

Category Weight Fail (<40%) Pass (40%-49%) Merit (50-59% Merit (60-69%) Distinction (70%+)
Investigation
Validation
Exercise
15%
Limited completion or
substantial inaccuracies;
weak evidence of progress.
Most checkpoints
addressed; sufficient
evidence of engagement
and basic analytical
accuracy.
Most checkpoints accurate,
with evidence of structured
investigation; some
validation or unexplained
errors remain.
Generally accurate
checkpoints with evidence of
systematic investigation and
validation.
Highly accurate and well-

validated findings; errors are identified, explained and corrected through investigation.

Investigation Report 45%

Incomplete, unclear or largely descriptive; weak methodology and reasoning.

Coherent report with appropriate methodology, analysis and findings addressing the case study. Brief AI use logs in investigation report writing.

Clear, well-organised report with sound methodology and evidence-based findings; uncertainty and incident context are partly addressed. Elaborate AI use methodology provided for report writing.

Well-structured analysis with clear technical reasoning, interpretation and incident context. Evidence of only free tier frontier AI use for coursework for report writing.

Professional, precise and evidence-led investigation with strong reasoning, critical interpretation and awareness of uncertainty. Evidence of only open-source AI use for report writing.

Technical Evidence 25%

Little or irrelevant evidence;

				evidence does not support conclusions.
				Relevant evidence supports key findings and demonstrates practical engagement. Brief AI use logs in technical evidence curation.
				Relevant evidence supports most findings with some annotation and linkage to conclusions; selection or reproducibility is not well elaborated. Elaborate of AI methodology in technical evidence curation.
				Good selection and integration of technical evidence; artefacts are clearly connected to findings. Evidence of only free tier frontier AI use for coursework for report writing.
				Comprehensive, well-selected and well-annotated evidence demonstrating authentic, reproducible investigation. Use of AI for technical evidence curation limited to only open source models.
Professional Practice	15%			Limited SOC awareness, incident-response reasoning or professional communication; AI reflection absent or superficial.
				Adequate SOC/IR context, clear communication and a

basic critical AI reflection.
Brief AI use for professionalism.
Sound SOC/IR context and clear communication; AI reflection covers use, benefits, some validation and limitations with good elaboration provided.
Good professional judgement, incident-response reasoning, communication and meaningful AI reflection using only free-tier models.
Strong SOC awareness, defensible incident-response reasoning, professional briefing and critical reflection on responsible AI use and limited to only open source models.

Table 5: CW1 Marking Rubrics

Distinction-level work should provide a report short and simple enough for executives to skim but not lacking the detail that tech leads will require. It should also include both strategical and operational recommendations to avoid future incidents based on standards (ISO, NIST, NCSC) documents.

** end of COMP3010 Module Handbook 2026-2027 - V2.pdf **

** Report.pdf full content **

COMP3010: Intrusion Analysis Investigation Report

1. Methodology Using Denning's Generic Intrusion Detection Model (ref), the intrusion incident was investigated on Ubuntu Linux by correlating Snort and Suricata alerts with Wireshark packets. This methodology was selected because Snort and Suricata's different preprocessors and rulesets reduce missed detections during triage, while Wireshark's packet decoding helps validate alerts and test hypotheses.

1.1 Triaging Alerts

CW1Snort.txt and CW1Suricata.txt were triaged in the Ubuntu terminal using 'grep', 'wc -l', and 'gedit' to rapidly find patterns in long logs and filter out false alarms. CW1Snort.txt was

filtered for Priority: 1 classifications (app -1) to prioritise severe threats and identify the internal host IP triggering alerts, while 'wc -l' was used on CW1Suricata.txt to quantify trojan check-ins (app0).

1.2 Wireshark
COMP3010CW1.pcap was opened in Wireshark on Windows and time display set to UTC Date and Time of Day to standardise timestamps across tools and avoid timezone differences.
Display filters were applied to isolate DHCP traffic for host identification. HTTP requests filter (http.request) (App1) was used to establish the infection timeline and inspect TCP streams (Follow -> TCP Stream) (App2).

1.3 File Extraction and Hashing
In Wireshark, HTTP object list (File -> Export Objects -> HTTP) (App3) was used to extract files sent over the network because some triggered IDS alerts (app), and have binary HTTP content types.

Linux ‘file’ command was used to confirm file types and ‘shasum -a 256’ to obtain SHA256 hashes (app4) for malware verification without executing the files.

1.4 Open-Source Intelligence (OSINT) Correlation To find the file that initially infected the computer, the SHA256 hash of gerv.gun was
searched on VirusTotal as it was the first file requested following the infection.

2. Analysis and Findings
   2.1. Infected System
   Correlating IDS logs (app5, 5.1.2, 5.1.3) confirms that 192.168.1.96 is the system compromised by a trojan downloader and spambot, creating immediate risks to Confidentiality (potential credential and data exfiltration), Integrity (unauthorised binary execution), and Availability (outbound spam blacklisting). Filtering for and analysing DHCP packets in Wireshark provides the host identification details: the IPv4 address is 192.168.1.96, the MAC address is 00:15:c5:de:c7:3b, and the hostname is FlashGordon-PC.

The Suricata policy alerts (app5.1.6), user-agent in Wireshark (app7), and the ‘file’ command output confirming that the payloads are Windows executables (app4) confirms that the operating system of FlashGordon-PC is Microsoft Windows.

2.2 How the Infection Occured
Wireshark and Snort log the infection on 27 June 2017 at 13:38 UTC (app8), whereas Suricata timestamps the first alert (Event#3.49002) on 28 June at 00:14:33 (app9). Matching IPs 119.28.70.207 and 192.168.1.96 proved Event#3.49002 is the same 13:38 UTC packet (Packet 204), correcting Suricata’s clock and establishing that the infection started on 2017-06-27 at 13:38 UTC.

Correlating the Wireshark TCP stream user-agent (app 7) with Suricata Event#3.49002 (app5.1.1) indicates Windows Script Host executed a script on FlashGordon-PC to download gerv.gun. Because the capture begins with this script requesting gerv.gun, it ruled out the hypothesis that the script would appear in Wireshark's HTTP objects.

To find the file that initially caused the infection, gerv.gun's SHA256 hash was searched on VirusTotal. A VirusTotal Community comment links an article (ref) detailing a relevant malspam campaign.

The article mentions two filenames:

These files were then searched in VirusTotal and their SHA256 hashes were found to be 59727ea201d94d471bdbb353b1281cca3c5bebf5305b140a3c017e52a37be702 and 64ae13592c5d8b289be81ebb50e054cef49b0bdf50ea92dd44ed611dd274150f. It is likely that one of these files was run on FlashGordon-PC which then called WinHttpRequest to get gerv.gun.

2.3 Identifying the Type of Infection
Based on the Snort and Suricata alerts, FlashGordon-PC was infected with Pushdo and spambot malware (Win.Trojan.Pushdo / Backdoor.Win32.Pushdo.s) (app5.1.2 & 5.1.3).

HTTP object list in Wireshark shows three executable files sent over HTTP during the infection.

These downloads triggered executable file alerts in Suricata and Snort (app5.1.1).

The three files were saved and verified using 'file' and 'shasum -a 256' (app4).
All files returned as 32-bit Microsoft Windows executables and each has the following hashes:

File name SHA256 hash
gerv.gun 0931537889c35226d00ed26962ecacb140521394279eb2ade7e9d2afcf1a7272
trow.exe 94a0a09ee6a21526ac34d41eabf4ba603e9a30c26e6a1dc072ff45749dfb1fe1
wp.exe 79d503165d32176842fe386d96c04fb70f6ce1c8a485837957849297e625ea48

Following the execution of these files, Suricata recorded 823 alerts (app0) for Pushdo trojan and many check-ins (app5.1.2). Snort logged Pushdo check-ins (app5.1.3). Several check-ins contacted sinkholes, triggering Suricata sinkhole alerts (app5.1.4). Additionally, Snort recorded alerts for SMTP buffer overflow attempts (app5.1.5), showing command-and-control check-ins and spam distribution.

3. Technical Reasoning and Uncertainty
   Observed facts verified from the PCAP and IDS logs are that FlashGordon-PC requested gerv.gun at 13:38 UTC on 27 June 2017, downloaded the gerv.gun, trow.exe, wp.exe

executables, and generated over 823 Pushdo check-in alerts plus SMTP traffic. There is evidence to suggest that the compromise could have originated from a malspam campaign containing an obfuscated .js file. Because the network-based IDS sensors lack host-level visibility, the .js file itself is not in the PCAP, and its local execution is inferred from the WinHttp.WinHttpRequest.5 User-Agent and the OSINT evidence from gerv.gun's SHA256 hash on VirusTotal to myonlinesecurity.co.uk. Suricata sinkhole alerts (app5.1.4) show some Pushdo check-ins were redirected, but others still delivered trow.exe and wp.exe, proving the PC remained compromised. Finally, uncertainty remains because no host-based IDS sensors or email logs are available, so it cannot be confirmed which exact .js hash (59727ea2... or 64ae1359...) was executed, which user account opened the attachment, what persistence mechanisms were established, or if data was exfiltrated.

4. Operational and Strategic Recommendations
   The following controls are recommended to contain the incident and prevent recurrence in alignment with ISO 27001 and NCSC and NIST guidance. At the operational level (NIST ref), the Security Operations Centre should immediately isolate FlashGordon-PC from the network, block the malicious payload and the command-and-control IP addresses (119.28.70.207, 145.131.10.21, and 96.82.200.1) in the firewall; and, to prevent spambots from propagating, restrict outbound SMTP traffic. Forensic images should be taken from FlashGordon-PC to recover the initial infection file. At the strategic level (ISO, NCSC ref), the organisation should deploy host-based IDS to close the gap in endpoint visibility and automate host isolation. Outdated software flagged by Suricata (Internet Explorer 6) should be decommissioned. Finally, because the files that potentially started the infection were obfuscated by using double extensions (DOC0978043-2017.6.doc.js and 2017-547805549-231.pdf.js), Windows File Explorer should be configured via Group Policy to display the file extension. Additionally, employees should be trained (ISO 27001, NCSC) to verify that files are the type they expect before opening them.

References

App-1

App0

App1

App2 Wireshark TCP Stream (Follow -> TCP Stream)

App3 Wireshark's HTTP object list (File -> Export Objects -> HTTP)

App4

App5

5.1.1 Suricata executable alerts

5.1.1.2 Snort Executable alerts

5.1.2 Suricata Pushdo check-ins

5.1.3 Snort Pushdo check-ins

Note: checkin from infected host to comm-and-ctrl server
5.1.4 Suricata sinkhole connection attempts and replies

5.1.5 Snort SMTP buffer overflow suggesting infected host is sending spam emails

5.1.6 Example of Suricata 'Internet Explorer 6 in use' alert indicating that the host runs Windows operating system.

App7 User-agent requesting gerv.gun contains 'Win32; WinHttp.WinHttpRequest.5' showing a script was run on Windows

App8
Wireshark's first log on 27 June 2017 at 13:38 UTC

Snort's first log on 27 June 2017 at 13:38 UTC

App9
Suricata's first alert on 28 June at 00:14:33

** end of Report.pdf **

Based on the content above, please provide a response to the user query.

User query: Using the Report document and the brief, write the script for a first class video briefing

Assistant

Video Briefing Script: Incident Analysis Report

(Duration: 5 minutes)

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1. Introduction (0:00–0:30)

"Good morning. My name is [Your Name], and I'm presenting on an intrusion incident detected in our network on June 27, 2017. This briefing summarizes the investigation findings, technical rationale, and actionable recommendations to mitigate risks. Our goal is to inform leadership of the threat level, explain how we identified and contained it, and outline steps to prevent recurrence."

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2. Incident Overview (0:30–2:00)

*“We detected a compromise on FlashGordon-PC (IP: 192.168.1.96), an internal workstation running Microsoft Windows. The system was infected with Pushdo trojan and spambot malware via a malspam campaign involving obfuscated files like `gerv.gun`. Initial detection occurred on June 27 at 13:38 UTC, with subsequent C2 check-ins and outbound spam attempts.

Key findings:

- **Infection method**: The system downloaded malicious payloads after executing a script triggered by a phishing email containing a double-extension file (`DOC0978043-2017.6.doc.js`). This bypassed basic antivirus and web filters.
- **Malware type**: Pushdo trojan (`Backdoor.Win32.Pushdo.s`) with spambot capabilities, evidenced by 823 C2 alerts, SMTP buffer overflow attempts, and sinkhole redirects.
- **Impact**: Confidentiality risks (credential exfiltration), integrity threats (unauthorized binary execution), and availability impacts (outbound spam blacklisting).

Technical validation: Correlated Snort/Suricata logs with Wireshark packet captures to confirm timestamps, IPs, and file hashes. Open-source intelligence (VirusTotal) linked `gerv.gun` to a known malspam campaign.”*

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3. Investigation Methodology & Evidence (2:00–3:15)

*“Our approach aligned with NIST SP 800-61 and ISO 27001 standards. We used open-source tools like Wireshark, Snort, and Suricata to triage alerts and validate findings.

Critical steps:

1. **Alert triage**: Filtered logs for high-priority detections (e.g., Pushdo check-ins) using grep/wc commands.
2. **Packet analysis**: Used Wireshark’s HTTP object list to extract payloads (`gerv.gun` , `trow.exe` , `wp.exe`) and cross-referenced SHA-256 hashes with VirusTotal.
3. **Host identification**: DHCP logs confirmed FlashGordon-PC’s MAC/hostname, while Suricata alerts indicated outdated software (Internet Explorer 6).
4. **Timeline reconstruction**: Matched timestamps across tools to correct clock discrepancies and establish the initial infection window.

Uncertainties noted: No host-based IDS or email logs were available, so the exact phishing attachment hash and user account remain unknown. However, network-based evidence strongly supports a malspam-driven compromise.”*

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4. Immediate Actions & Recommendations (3:15–4:30)

*“To contain the incident and prevent recurrence, we recommend:

Operational controls:

- Isolate FlashGordon-PC from the network immediately. Block malicious IPs (119.28.70.207, 145.131.10.21, 96.82.200.1) and restrict outbound SMTP traffic.
- Preserve forensic images for malware analysis.

Strategic controls:

- Deploy host-based IDS to close endpoint visibility gaps (per ISO 27001).
- Update Group Policy to display file extensions to deter obfuscated files.
- Train users on phishing awareness, referencing NCSC guidelines.
- Decommission outdated software (e.g., Internet Explorer 6) per Suricata alerts.

Alignment with standards: Recommendations follow NIST SP 800-61 for incident response and ISO 27001 for risk mitigation.”*

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5. Conclusion & Next Steps (4:30–5:00)

“*This incident highlights the importance of layered defenses, including endpoint visibility and user training. We’ve validated findings through open-source tools and OSINT, ensuring accuracy in our response. For further details or questions, please contact [SOC Lead/Incident Manager]. Thank you for your attention.*”

Key References:

- NIST SP 800-61 (Incident Response Framework)
- ISO 27001 (Information Security Management)
- NCSC Guidance on Malspam Defense
- VirusTotal OSINT Correlation

(End of Script)